

Popular design areas

Networking. Lots of opportunities in fibre optics

Processors. Ultra-fast processors, low response times, speed above 2 GHz, multiprocessor chips, two-processor system-on-chips (SoC), dual-core and quad-core, video and networking applications

Automotive. Significant growth (from 8 per cent to 20 per cent) in electric cars, and increased penetration of electronics in automobiles. Small cars will also support electronics, so there are tremendous opportunities in the automotive space. Balaji points out some important areas: "Sophisticated engines and electronics are used to reduce carbon emission with the new standards coming up. Automotive safety is also gaining importance—connectivity and navigation systems play an important role in this. MP3 players and other entertainment devices are also being designed."

Embedded. Embedded software is gaining dominance. "Mixed-signal processors are becoming more complex and SoC integration is becoming a must," says Mehtani.

Next-generation products

1. Multi-point and multi-touch touch-screen controllers
2. Designing for larger screen sizes
3. Lowering power for various applications
4. Capsense switches: In automotive applications like dashboards, MP3 players, cellphones, washing machines, etc
5. Engine controls for e-bikes
6. Projects related to power devices
7. Balancing renewable energy systems
8. Internet protocol multimedia subsystem on satellite technology
9. Alternative fuel-powered systems
10. Instrument cluster with colour TFT screen
11. Touch-screen-based navigation

sales, India, Australia & New Zealand, Xilinx Development Corporation, shared similar goals: "We believe that building compelling products that capitalise on the trend toward programmability will continue to enable strong and steady growth for Xilinx." He added, "The cycle of technology is in our favour, the economics are in our favour, and we're committed to accelerating the 'programmable imperative' by leading the industry in silicon innovation on the world's most advanced chip-making technology and delivering simpler, smarter, standards-based programmable 'targeted design platforms' that free up designers to focus on end product innovation and differentiation."

However, creating the best designs is no mean task. One has to keep up

with the innovation imperative, while also meeting the increasingly stringent quality norms and technical requirements—which range from reducing power consumption and cost to being eco-friendly. But, IDCs are undeterred by these challenges.

"Our R&D engineers constantly go through training and orientation programmes on the latest techniques and tools used in semiconductor and VLSI design, architectures, EDA tools, embedded software, etc. We have a strong collaboration programme with our labs in other centres of TI and we share and discuss the latest developments in technology. Our leaders constantly mentor their team members and help them face new technical challenges," informed Mitra.

Customers are beginning to expect more and more. Instead of being scared by these demands, companies are leveraging this opportunity to co-create. As Mitra said, "We are converting this challenge into an opportunity by moving closer to our customers through our sales and applications support offices. We are co-innovating with our customers, speaking the customers' vocabulary and delivering within the stipulated time frame. We use simple strategies like listening, understanding and serving the needs of our customers. Our R&D engineers are always immersed in the customers' world. We are co-innovating with our customers. We are not just creating chips but innovating with solutions. We are passionate about solving our

customers' problems. For us, even small customers are important, as we believe that large customers were also small at some point of time."

Another strategy to maintain constant growth is to leverage the distributors and third-party design houses. Mahendra Jain, country manager, Actel India, shared, "We plan to leverage our distributors, as well as work closer with design houses that support multinational companies in consumer and industrial markets."

Actel even tries to build an excellent rapport with students, as that provides an opportunity to tap tomorrow's market. "We are also very much involved in the universities, teaching engineering students all about our products and tools. When they find jobs in India or any other place in the world, they will already have a natural affinity and preference for Actel products," explained Jain.

Countering competition

The onus rests on design houses to make a product better than a competitor's. That is a tall order, but a servable one for IDCs, which follow systematic strategies to counter competition.

Raju Pudota, managing director, Denali Design Systems, said, "Tap the niche markets before your competitors do, so that you are the pioneer in that particular market. Next, work with standards bodies like Jetlag DDR2, PCI Express and USB so that you can track policy changes instantly and help your respective customers. Another important strategy is to be quick in decision-making, because time is very important in competitive markets."

Vivek Sharma, regional vice president-Greater China & South Asia-India Operations, and director of India Design Centres, STMicroelectronics, said that offering broad and technologically-rich product portfolios helped in countering competition.

Product differentiation also works as an effective strategy. Rajeev Mehtani, senior vice president-India operations, Cypress Semiconductor